

## Fall 2005

**Problem 1 (Fa05).** Let  $M$  be a compact metric space and let  $(U_i)_{i \in I}$  be an open cover of  $M$ . Show that there exists  $\varepsilon > 0$  such that, for all  $x, y \in M$ , if  $d(x, y) < \varepsilon$  then there is some  $j$  with both  $x$  and  $y$  in  $U_j$ .

**Problem 2 (Fa05).** Prove that if  $f(z) = P(z)/Q(z)$  is a rational function with complex coefficients whose numerator has lower degree than the denominator, then  $f(z)$  is a sum of terms of the form  $a/(z-b)^k$ , with  $a, b \in \mathbb{C}$ .

**Problem 3 (Fa05).** Define  $U \subseteq \mathbb{C}$  to be the open right half-plane with the interval  $(0, 1] \subseteq \mathbb{R}$  deleted. Find an explicit conformal equivalence of  $U$  with the open unit disc  $\mathbb{D}$ .

**Problem 4 (Fa05).** Let  $m$  and  $n$  be positive integers. Prove that the ideal generated by  $x^m - 1$  and  $x^n - 1$  in  $\mathbb{Z}[x]$  is principal.

**Problem 5 (Fa05).** Is there a differentiable function  $f: \mathbb{R} \rightarrow \mathbb{R}$  satisfying  $f(0) = 1$  and  $f'(x) \geq f(x)^2$  for all  $x \in \mathbb{R}$ ?

**Problem 6 (Fa05, Fa18).** Let  $A$  be an  $n \times n$  matrix with real entries such that  $(A - I)^m = 0$  for some  $m \geq 1$ . Prove that there exists an  $n \times n$  matrix  $B$  with real entries such that  $B^2 = A$ .

**Problem 7 (Fa05).** Let  $f(z) = z^5 + 5z^3 + z^2 + z + 1$ . How many zeros (counting multiplicity) does  $f$  have in the annulus  $1 \leq |z| \leq 2$ ?

**Problem 8 (Fa05, Fa18).** Find the smallest  $n$  for which  $S_n$  contains a cyclic subgroup of order 111.

**Problem 9 (Fa05).** A doubly infinite sequence  $(a_j)_{j \in \mathbb{Z}}$  of real numbers is said to be *rapidly decreasing* if, for each positive integer  $n$ , the sequence  $j^n a_j$  is bounded. Let  $(a_j)$  and  $(b_j)$  be rapidly decreasing sequences, and define the convolution of these sequences by  $c_j = \sum_{k \in \mathbb{Z}} a_k b_{j-k}$  for  $j \in \mathbb{Z}$ . Prove that the series defining each  $c_j$  is convergent, and that  $(c_j)$  is a rapidly decreasing sequence.

**Problem 10 (Fa05).** How many pairs of integers  $(a, b)$  are there satisfying  $a \geq b \geq 0$  and  $a^2 + b^2 = 5 \cdot 17 \cdot 37$ ?

**Problem 11 (Fa05).** Let  $f$  be a continuous real valued function defined on  $[0, \infty)$ , such that  $f(x) \geq 0$ ,  $f$  is nonincreasing, and  $\lim_{x \rightarrow \infty} f(x) = 0$ . Show that

$$\lim_{R \rightarrow \infty} \int_0^R f(x) \sin x \, dx$$

exists.

**Problem 12 (Fa05).** For which pairs of monic polynomials  $(p(x), m(x))$  over the complex numbers does there exist a matrix in  $M_n(\mathbb{C})$  whose characteristic polynomial is  $p(x)$  and whose minimal polynomial is  $m(x)$ ?

**Problem 13 (Fa05).** Determine which numbers  $a \in \mathbb{C}$  have the following property: There exists an analytic function  $f$  defined in the open unit disc such that, for all integers  $n \geq 2$ ,  $f(1/n) = 1/(n+a)$ .

**Problem 14 (Fa05).** Given a prime number  $p$ , let  $\mathbb{F}_p$  be the field of  $p$  elements, and let  $R$  be the ring  $\mathbb{F}_p[x]/\langle x^3 \rangle$ . For which primes  $p$  is the unit group  $R^*$  cyclic?

**Problem 15 (Fa05, Sp14).** Let  $K \subset \mathbb{R}^n$  be closed, convex, and nonempty. Show that for every  $x \in \mathbb{R}^n$ , there exists  $y \in K$  that uniquely minimizes the Euclidean distance to  $x$ , i.e.,  $\|x - y\| < \|x - z\|$  for all  $z \in K \setminus \{y\}$ .

**Problem 16 (Fa05).** Let  $V$  be a finite dimensional complex vector space with a positive definite Hermitian inner product. Let  $T: V \rightarrow V$  be a Hermitian linear operator. Prove that

1.  $1 + iT$  is nonsingular;
2.  $(1 - iT)(1 + iT)^{-1}$  is a unitary operator.

**Problem 17 (Fa05).** For  $R > 0$  let  $\Gamma_R$  be the semicircle  $\{|z| = R, \Im z \geq 0\}$ . Prove that

$$\lim_{R \rightarrow \infty} \int_{\Gamma_R} \frac{e^{iz}}{z} \, dz = 0$$

**Problem 18 (Fa05).** Let  $f: \mathbb{R} \rightarrow \mathbb{R}$  be a function. Prove that there exists a countable subfield  $\mathbb{K}$  of  $\mathbb{R}$  such that  $f(\mathbb{K}) \subseteq \mathbb{K}$ .